Kaidi **Zha**

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"Building intelligent systems that see, learn, and act."

Research Interests

Generative Models, Inference Acceleration

Dedicated to making large generative models faster and cheaper to deploy through **quantization and mixed-precision inference**, **efficient architecture design**, and **data-centric strategies**, with a focus on **inference throughput**, **memory efficiency**, and **training efficiency**.

Education

Tsinghua University *Beijing, China*

B.S. in Mathematics and Physics *Sept. 2023 - Present (Expected 2027)*

- Overall GPA: 3.6/4.0, with a steady upward trend — 3.9/4.0 in the most recent semester (Spring 2026).
- Earned A (4.0) in Pattern Recognition and Machine Learning, Reinforcement Learning and Control, and Fundamentals of Computer Graphics.

Relevant Coursework

AI & ML	Pattern Recognition and Machine Learning (A), Reinforcement Learning and Control (A), Fundamentals of Computer Graphics (A), Statistical Machine Learning (B+)
Mathematics	Probability and Statistics (A), Experiments in Mathematics (A), Applied Stochastic Processes (A-), Complex Analysis (B+)
Computer Science	Data Structure and Algorithm (A-), Programming Fundamentals (A-), Modern Operating Systems (P), C++ Programming for Linux (P)

Research Experience

Q-DiT: Accurate Post-Training Quantization for Diffusion Transformers *Supervised by Prof. Yuan Meng, Dept. of CS & Tech, Tsinghua University*

Research Assistant *Jun. 2025 - Present*

- **Motivation**: Uniform-precision PTQ wastes the bit budget on diffusion Transformers, whose quantization sensitivity varies sharply across denoising timesteps and network layers — bits should be spent where the loss surface is steepest.
- **Method**: Co-designed a timestep-aware, layer-wise mixed-precision PTQ framework grounded in second-order analysis: the Hessian trace of the denoising loss scores each timestep's sensitivity, GPTQ (Optimal Brain Surgeon) error bounds per-layer weight perturbations, and Fisher information ranks each layer's contribution to the training objective. Their combination turns bit allocation into a constrained optimization problem, solved exactly by dynamic programming under a global W4A4 budget, with a cross-adjustment mechanism that rebalances weight and activation bit-widths.
- **System & Results**: Independently refactored the codebase into *qdiffusers*, a modular YAML-driven quantization–evaluation pipeline, and adapted it to **7 DiT-family models** (DiT-XL/2, FLUX.2, PixArt-Σ, OpenSora, Qwen-Image, HunyuanVideo, Wan2.2) with multi-GPU parallel GPTQ. Led all ImageNet-1K experiments — W4A4 achieves near-lossless FID 7.76 on DiT-XL/2 256×256; on FLUX.2-klein-4B, W4A4 FID 25.22 even **surpasses the FP16 baseline** (26.08).

ArchEGraph: A Large-Scale Graph Dataset for Geometry–Topology–Physics Aligned Building Energy Modeling *Supervised by Prof. Borong Lin, School of Architecture, Tsinghua Univ.; with UC Berkeley & MIT*

Research Assistant *Apr. 2026 - Present*

- **Role**: Second author of the manuscript; independently designed and executed the benchmark experiments for a dataset of **5,481 buildings and 49,326 validated simulation cases** ($>1.33\times10^5$ space nodes, 1.44×10^6 face nodes).
- **Contributions**: Built the evaluation protocols for two benchmark tasks — reconstructing zone topology from polygonal meshes (M2G) and topology-informed zone-level load prediction — and benchmarked 7 model families (XGBoost, DeepSets, Set Transformer, Perceiver, GATv2, TransformerConv, GraphGPS) under cross-building and cross-climate splits.
- **Findings**: Showed that graph-based surrogates learn the geometry–topology–physics coupling well in-distribution, while cross-climate and cross-layout generalization remains the key bottleneck — empirically validating the dataset's design goal of exposing geometric and climatic shift.

Selected Projects

TrainYourOwnGPT: Building & Analyzing GPT-Style Language Models from Scratch

Pattern Recognition & Machine Learning, Course Project

TEAM OF 2; LED SCALING-LAW AND FINE-TUNING STUDIES *Spring 2026*

- **Motivation**: Understanding why language models scale — and when architectural tweaks and adaptation genuinely help — requires controlled experiments built from first principles, not off-the-shelf pipelines.
- **Contribution**: Built the full stack from scratch: a GPT-style decoder (RoPE, SwiGLU, pre-norm blocks) trained on Penn Treebank, and led the scaling-law and fine-tuning studies (team of 2).
- **Method**: Ran a 5-configuration hyperparameter sweep × 3 seeds; studied scaling over 0.77M–5.46M parameters with controlled ablations of QK normalization, attention gating, and value embeddings; fine-tuned Qwen2.5-0.5B with LoRA (0.89% parameters) on arXiv abstracts under a pre-registered evaluation; compared autoregressive vs. masked-diffusion LMs (MDLM, LLaDA) with A100 throughput benchmarks.
- **Results**: **105.15 validation PPL** for the best scratch model; attention gating gives the most reliable architectural gain (-2.02 PPL); LoRA adaptation eliminates exam-style outputs (4/15 → 0/15) and yields 8× more abstract-style continuations.

fair-SPLIT: Near-Optimal Decision Trees for Accuracy, Sparsity, and Fairness

Statistical Machine Learning, Course Project

TEAM OF 2; DEVELOPED THE SPLIT-FAMILY METHODS AND EXPERIMENTAL DESIGN *Spring 2026*

- **Motivation**: Greedy tree induction (CART) is myopic and offers no fairness guarantee, while simply dropping sensitive attributes does not remove indirect discrimination — we studied how far near-optimal trees plus lightweight post-processing can close this gap.
- **Method & Experiments**: Implemented the SPLIT family, which replaces greedy induction with dynamic-programming lookahead that provably optimizes the most discriminative top layers before greedy completion; benchmarked against CART on **12 public datasets** (6 with sensitive attributes) × 5 seeds, measuring accuracy, training time, and SPD/DI/EOD fairness metrics — SPLIT variants win on COMPAS, HELOC, Heart, and Law School.
- **Fairness**: Proposed Leaf-Pareto Fair Recalibration (LPFR), a deployable leaf-level post-processing method; label-aware calibration cuts the demographic-parity gap from 0.17 to **0.003** on Adult. Open-sourced at github.com/illusionsin30/fair-SPLIT.

AIDER: Taming Aleatoric Impulse in Off-Policy Reinforcement Learning

Reinforcement Learning & Control, Course Project

TEAM OF 8; EXPLORED DIFFUSION/FLOW-BASED MULTIMODAL POLICY EXTENSION *Spring 2026*

- **Motivation**: Off-policy RL suffers a transient value-overestimation surge (the *aleatoric impulse*) rooted in intrinsic estimation noise; existing heuristic annealing schedules are hyperparameter-sensitive and asymptotically inconsistent.
- **Contribution**: The team proposed DEAR, an EMA-anchored evidential regression operator that learns higher-order aleatoric uncertainty end-to-end, and AIDER, which turns it into pessimistic targets and exploration bonuses with no annealing schedule; I explored extending the framework to diffusion / normalizing-flow multimodal policies.
- **Method**: A higher-order EMA target network serves as an empirical anchor, and a Gaussian NLL objective lets each critic directly estimate its own target-value variance through an evidence-count parameterization; total value uncertainty (ensemble epistemic + learned aleatoric) drives both a lower-confidence-bound critic target and an upper-confidence-bound behavior policy.
- **Results**: Best average performance on the DeepMind Control Suite among 7 model-free baselines (Humanoid-run return 575 → 648 over the predecessor); deployed on a 4WD vehicle speed-tracking task with **RMSE 0.349 m/s** (∼1.5% error).

CPU Path Tracer: A Physically-Based Renderer in C++

Fundamentals of Computer Graphics, Course Project

INDIVIDUAL PROJECT *Spring 2026*

- **Motivation**: Physically-based rendering is Monte Carlo estimation of the rendering equation — naive sampling converges slowly, so variance reduction and acceleration structures are what make it practical.
- **Contribution**: Independently built a complete CPU renderer in C++ bridging Whitted ray tracing and Monte Carlo path tracing.
- **Method**: Next-event estimation for direct lighting, Cook–Torrance glossy BRDF with Schlick Fresnel, Russian-roulette path termination, BVH acceleration, and depth of field; engineered deterministic per-pixel seeding with lock-free row dispatch for multithreading.
- **Results**: **49× speedup on 64 threads** with byte-identical output to the serial renderer; every effect validated by controlled ablation renders (NEE on/off, Fresnel, roughness sweep, total internal reflection).

Skills

Programming	Python, C/C++, CUDA, LaTeX
Deep Learning	PyTorch, JAX, HuggingFace Transformers & Diffusers, PEFT/LoRA, quantization toolchains
Tools	Git, Linux, Docker, multi-GPU training, OpenMP / `std::thread`
Languages	Chinese (Native), English (CET-4: 596, CET-6: 534)